Student Assignment Brief

CONFIDENTIAL DOCUMENT

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Assignment Information

Module Name:	Artificial Intelligence
Module Code:	ST5001CMD
Assignment Title:	ST5001CMD Artificial Intelligence March Intake 2026 Coursework ST5001CMD (Regular)
Assignment Due:	28-July-2026, 11:55PM

Assignment Credit:	20
Word Count:	2000
Assignment Type:	Coursework : Individual Report
Grading:	Percentage Grade (Applied Core Assessment).

Assessment Overview

You will be provided with an overall grade between 0% and 100%. You have one opportunity to pass the assignment at or above 40%.

Important Notice

The work you submit for this assignment must be your **own independent work**. More information is available in the 'Assignment Task' section of this assignment brief.

Assessed Module Learning Outcomes

The Learning Outcomes for this module align with the marking criteria which can be found at the end of this brief. Ensure you understand the marking criteria to ensure successful achievement of the assessment task. The following module learning outcomes are assessed in this task

1. Evaluate the social and legal implications of artificial intelligence.
2. Articulate the various approaches to knowledge representation in artificial intelligence.
3. Explore reasoning and problems solving approach under uncertainty conditions.
4. Demonstrate an understanding of knowledge-based systems, and their respective architectures.
5. Resolve which of the artificial intelligence approaches offer the best solution to an intelligent application.
6. Evaluate human factors and the impact on Artificial Intelligence.

Assignment Task

Overview

This individual coursework requires you to design, implement, and critically evaluate an AI-driven solution to a real-world problem. Unlike a simple machine learning implementation, this assessment spans the full breadth of Artificial Intelligence. You must demonstrate knowledge across multiple AI domains, address social and legal considerations, and produce a reflective analysis of how your chosen approach compares with alternatives.

The assessment is structured into five interconnected parts, each of which targets specific module learning outcomes. Completing all five parts is mandatory.

Part A — Problem Definition and Domain Context (10 marks)
Select a real-world problem that lends itself to an AI solution. Your problem must be approved by your module leader before you proceed.
You must produce:

- A clear, well-scoped problem statement with defined inputs, outputs, and constraints.
- A justification of why AI is the appropriate solution paradigm for this problem.
- An identification of the AI problem domain(s) your solution will draw upon (see Section 4).
- An explanation of the real-world significance of the problem — who is affected, why it matters.
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Part B — AI Survey and Comparative Knowledge Representation Analysis (20 marks)
This part directly assesses LO2 and LO4. You must demonstrate understanding of the breadth of AI — not just the technique you have chosen to implement.
You must produce:

- A structured survey of at least three distinct AI approaches relevant to your chosen problem domain (e.g., rule-based systems, search algorithms, machine learning, fuzzy logic, genetic algorithms).
- For each approach, describe: (a) how knowledge is represented, (b) the system architecture, and (c) its suitability for your problem.
- A comparative analysis table evaluating the approaches against defined criteria (accuracy, interpretability, scalability, uncertainty handling).
- A justified conclusion identifying which approach you will implement, and why the others were discarded.

Note: Simply describing your chosen algorithm is insufficient for this part. You must engage critically with multiple AI paradigms.

Part C — Design and Implementation (30 marks)
Implement your chosen AI solution. Your implementation must go beyond a standard machine learning pipeline.
Requirements:

- Your solution must be implemented from a principled AI design — custom data structures, state representations, or knowledge bases must be defined where

appropriate.

- At least one component of your solution must address reasoning or decision-making under uncertainty (e.g., probabilistic reasoning, fuzzy membership, heuristic search, constraint propagation).
- You must include a system architecture diagram showing how components interact.
- Code must be version-controlled using GitHub Classroom with frequent, meaningful commits.
- A working interface (CLI, GUI, or simulation/visualisation) must be developed.

Part D — Evaluation and Experimental Results (20 marks)
Evaluate your solution rigorously. This part assesses LO5 — whether you can justify that your AI approach is the best solution.
You must:

- Define and apply appropriate performance metrics for your problem type.
- Run a minimum of three experimental configurations or test scenarios and report results.
- Compare your implemented approach against at least one baseline or alternative method.
- Reflect critically on the limitations of your solution and under what conditions it fails.
- Discuss the usability of your interface and how design choices affect the user experience.

Part E — Critical Reflection: Ethics, Law, and Human Factors (20 marks)
This part explicitly assesses LO1 and LO6, which are not covered by implementation alone.
You must produce a 600–800-word critical reflection addressing all three of the following:

- Social and Legal Implications (LO1): Identify at least two specific legal or regulatory frameworks relevant to your AI system (e.g., GDPR, EU AI Act, algorithmic fairness legislation). Discuss potential harms, biases, or risks your system could introduce and how these might be mitigated.
- Human Factors (LO6): Analyse how human behaviour, cognitive biases, or organisational contexts affect how your AI system would be used in practice. Discuss trust, explainability, and the human-in-the-loop considerations for your system.
- Future Improvements: Propose two concrete improvements that would better address either ethical concerns or human usability.

Eligible AI Problem Domains
Your implementation (Part C) must draw primarily from one of the following AI domains. Your survey (Part B) must engage with the others as alternatives:

- Search techniques (informed/uninformed, adversarial search)
- Constraint satisfaction problems and game playing
- Recommendation systems
- Classification and clustering
- Genetic algorithms and fuzzy logic systems
- Knowledge-based systems and expert systems

Choosing a pure machine learning classification or regression task without engaging with knowledge representation, reasoning under uncertainty, or system architecture will not satisfy the learning outcomes of this module.

Deliverables

- GitHub Repository: All source code with meaningful commit history. Submit the repository URL.
- Written Report (4000 words excluding references): Must cover all five parts. Use APA 7th referencing throughout.
- Video Demonstration (10 minutes): Problem overview, code and interface walkthrough, model in action, results explanation, and a 2-minute reflection on ethical/legal considerations.

Submission Instructions

Requirement	Details
File Naming	NAME_studentID
File Format	.docx/.pdf format
Submission Method	Campus 4.0 platform (submission link provided 2 weeks before deadline)

Marking and Feedback

How will my assignment be marked?
Your assignment will be marked by the Module Team using standardized criteria.

How will I receive grades and feedback?

Provisional marks will be released once internally moderated. Feedback will be provided alongside grades release within 2 weeks (10 working days).

What will I be marked against?

Details of the marking criteria for this task can be found in the Assessment Marking Criteria section at the end of this brief.

Grade Requirements

You must achieve 40% or above to pass this assessment. Ensure you understand the marking criteria for successful completion.

Assignment Support and Academic Integrity

Getting Help

If you have any questions about this assignment, please meet your respective module leader or teacher for more information.

Language Standards

You are expected to use effective, accurate, and appropriate language within this assessment task.

Academic Integrity

The work you submit must be your own. All sources of information need to be acknowledged and attributed; therefore, you must provide references for all sources of information and acknowledge any tools used in the production of your work, **excluding Artificial Intelligence (AI)**. We use detection software and make routine checks for evidence of academic misconduct. Definitions of academic misconduct, including plagiarism, self-plagiarism, and collusion can be found in Student handbook in Campus 4.0. All cases of suspected academic misconduct are referred to for investigation, the outcomes of which can have profound consequences to your studies.

Support for Students with Disabilities

If you have a disability, long-term health condition, specific learning difference, mental health diagnosis or symptoms, contact the Student Support Office for assistance.

Unable to Submit on Time?

If events prevent you from submitting on time, guidance on extenuating circumstances is available in the Student Handbook or from the Student Support Office.

Administration of Assessment

Module Leader Name:	Suman Shrestha
Module Leader Email:	stw0023@softwarica.edu.np
Assignment Category:	online
Attempt Type:	regular
Component Code:	ST5001CMD

Assessment Marking Criteria

Component	Description	Marks (%)
Part A: Problem Definition	Clear, scoped problem with justified AI relevance and domain identification.	10
Part B: AI Survey & Comparison	Breadth of AI knowledge, quality of comparative analysis, justified algorithm selection.	20
Part C: Implementation	Design quality, use of appropriate AI technique, uncertainty handling, architecture, interface, GitHub commits.	30
Part D: Evaluation	Rigour of experiments, use of metrics, comparison with baseline, usability reflection.	20
Part E: Ethics, Law & Human Factor	Depth of legal/regulatory engagement, human factors analysis, quality of reflection.	20
Total		100

	40% >=above	50% >= above	60%>=above	70% >= above
Problem Understanding and Dataset Handling (15%)	Weak problem definition with little justification. Uses an inappropriate or poorly prepared dataset, with significant preprocessing issues.	Basic problem definition with limited justification. Uses a dataset but preprocessing is minimal or has noticeable flaws.	Defines the problem well with good justification. Uses a relevant dataset with proper preprocessing but minor issues	Clearly defines the problem with strong justification. Uses a high-quality dataset, applies advanced preprocessing techniques, and handles missing/noisy data effectively.

Model Selection and Implementation(20%)	Model selection is weak or inappropriate. Implementation is functional but poorly structured, inefficient, or missing key elements.	Model selection is acceptable but lacks proper justification. Implementation works but has inefficiencies or lacks clarity.	Selects a suitable model with good justification. Code is mostly efficient with minor issues. Some parameter tuning is attempted.	Selects the best-suited model with strong justification. Implements efficiently with clean, well-documented code. Consider hyperparameter tuning.
Performance Evaluation & Optimization (20%)	Uses incorrect or minimal evaluation metrics. No meaningful optimization attempted.	Uses basic evaluation metrics but lacks deeper analysis or optimization.	Uses proper metrics and applies some model evaluation techniques. Minor optimizations attempted.	Uses appropriate evaluation metrics, applies cross-validation, and compares models. Optimizes performance effectively.
Innovation and real-world applicability (15%)	Lacks innovation. Minimal real-world applicability considered.	Project follows standard approaches with little innovation. Application to real-world scenarios is weak.	Shows some innovation and attempts to apply practical improvements.	Demonstrates creative thinking, applies novel techniques, and considers real-world applications effectively.
Presentation and Documentation (20%)	Report is disorganized, missing key explanations. Presentation lacks clarity or engagement.	Report is understandable but lacks depth, clarity, or proper formatting. Presentation is average.	Good report structure with mostly clear explanations. Presentation is organized but lacks some details.	Clear, well-structured report with strong justification and references. Presentation is engaging, logical, and professional.

This document is confidential and intended solely for enrolled students.